Genuine Multipartite Entanglement Analysis

```
Model
-----
Hubbard ring (1-D, PBC)

System parameters
-----------------
Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 10.0
Interaction (U)   : 1.0
Mode              : uniform
n-body order      : 4
```

```
Ground-state properties
-----------------------
E0                : -39.26003999
E1                : -39.25394536

<S2>              : 0.000000
S                 : 0.0000
<Sz>              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
<D> total         : 0.730035
<D> / site        : 0.182509
```

```
GME summary
-----------
Extracted n-spin RDM : n = 4

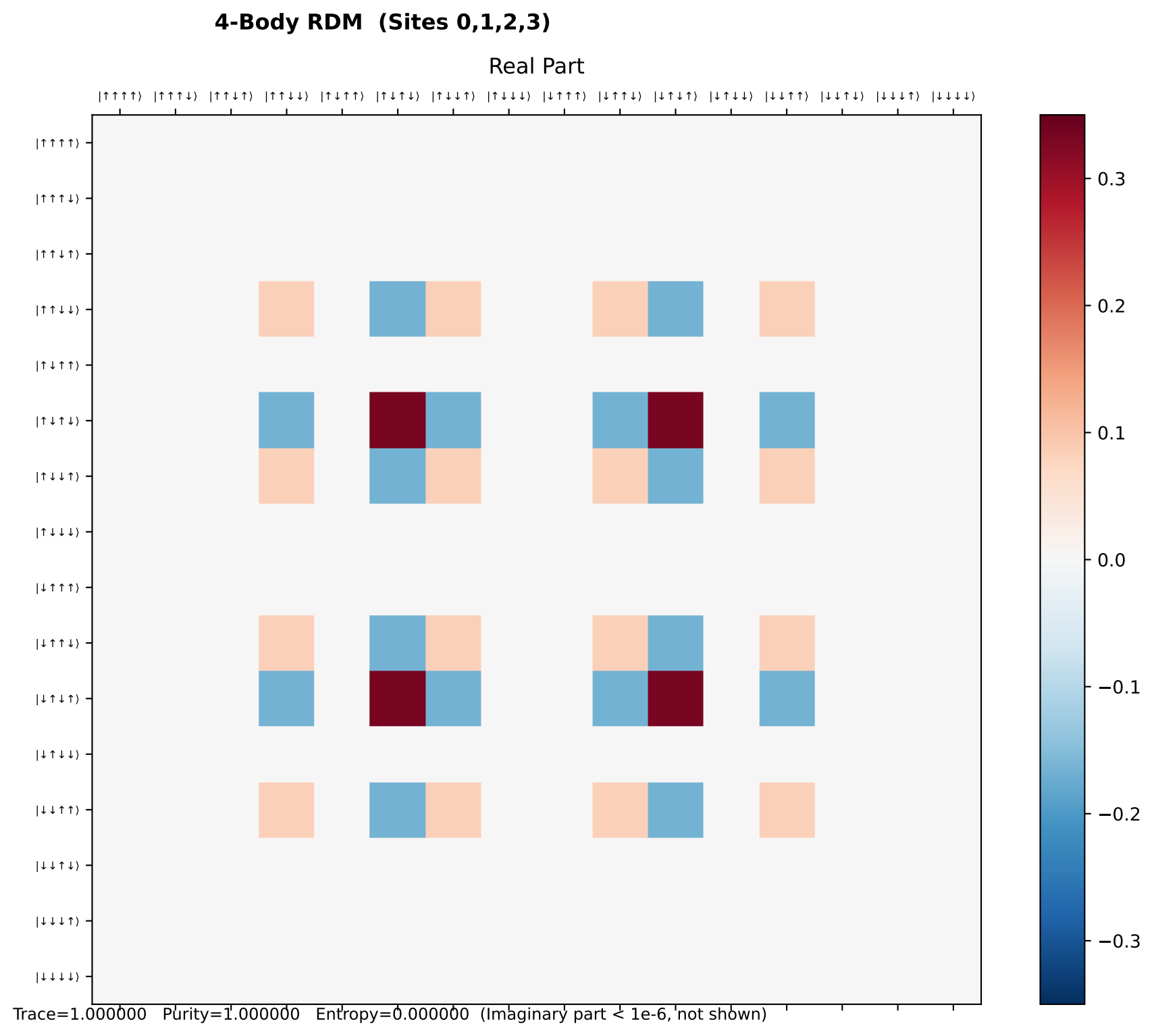
Purity:
  gamma = Tr(rho^2)

Separable cut criterion:
  gamma_A = 1  <->  rho = rho_A x rho_B

C_GME (Troiani et al.):
  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Decision logic:
  gamma=1, all Tr(rho_A^2)<1  -> GME
  gamma=1, some Tr(rho_A^2)=1 -> No GME
  gamma < 1                   -> Mixed

Date: 2026-04-01  14:49
```


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.985001e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3)	0.50000000
(0, 1)	(2, 3)	0.58333333
(0, 2)	(1, 3)	0.33333333
(0, 3)	(1, 2)	0.58333333
(0, 1, 2)	(3,)	0.50000000
(0, 1, 3)	(2,)	0.50000000
(0, 2, 3)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME (exact eq.1) = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.
lower bound [eqs.6-9]

C_LB = 0.66666667
Gap = 0.24620426 (C_exact - C_LB)
Best z1 = [0, 1, 0, 1]
Best z2 = [1, 0, 1, 0]

Continuous Optimization
==========

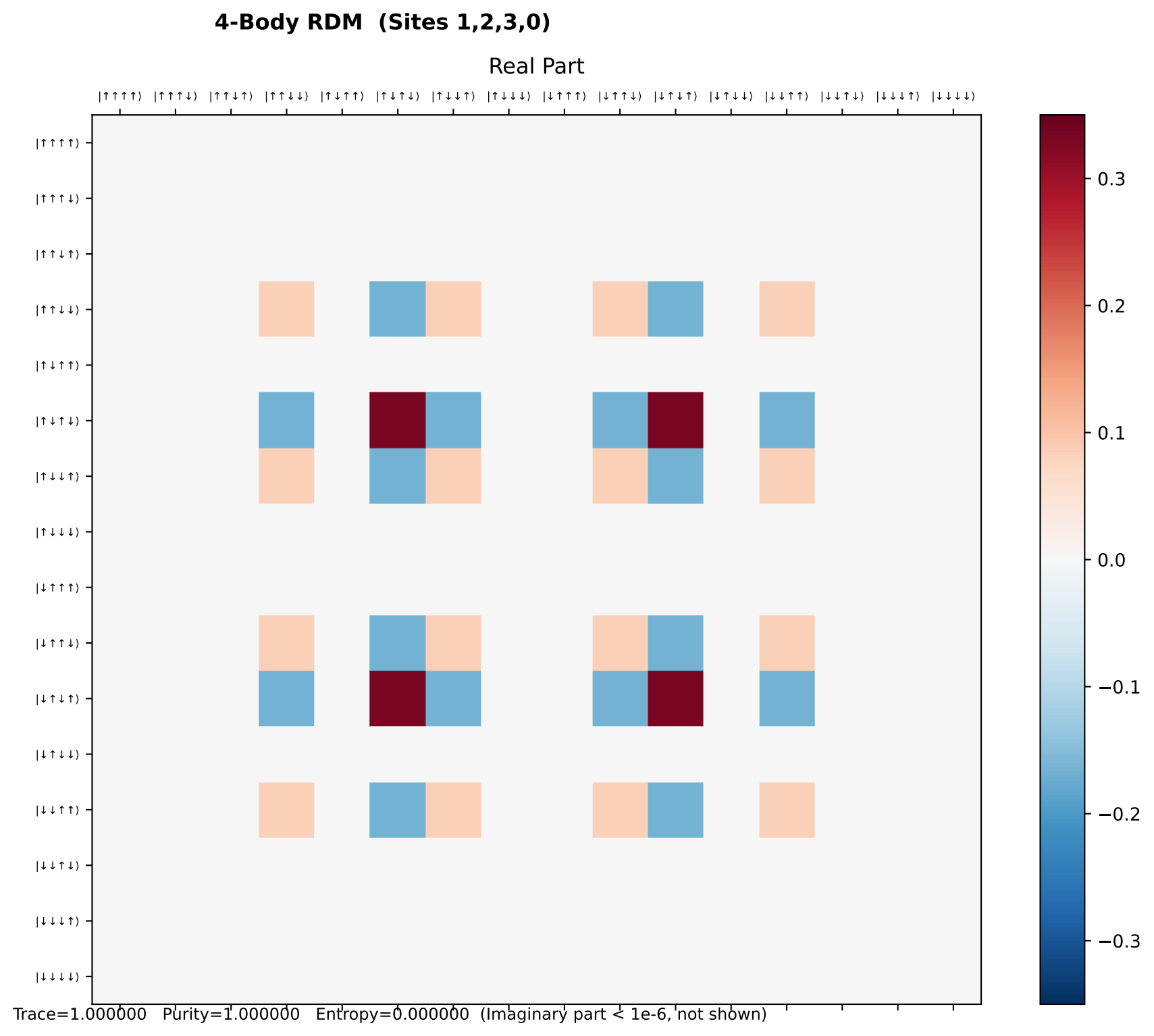
Each spin probed as a general superposition
|phi_k> = cos(t/2)|up> + exp(ip)*sin(t/2)|down>

Method	continuous_complex
C_LB (basis)	0.66666667 <- discrete search
C_LB (cont.)	0.66666667 <- continuous search
Gain	-0.00000000

Continuous Optimization Details
==========

Parameters per spin: theta [0,pi], phi [0,2pi]
Total search space : 4 x n = 16 parameters

Best thetas psi1:
[3.142, 0.0, 3.142, 0.0]
Best thetas psi2:
[0.0, 3.142, 0.0, 3.142]


4-Body RDM (Sites 1,2,3,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.589849e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 0)	0.50000000
(1, 2)	(3, 0)	0.58333333
(1, 3)	(2, 0)	0.33333333
(1, 0)	(2, 3)	0.58333333
(1, 2, 3)	(0,)	0.50000000
(1, 2, 0)	(3,)	0.50000000
(1, 3, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME (exact eq.1) = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.
lower bound [eqs.6-9]

C_LB = 0.66666667
Gap = 0.24620426 (C_exact - C_LB)
Best z1 = [0, 1, 0, 1]
Best z2 = [1, 0, 1, 0]

Continuous Optimization
==========

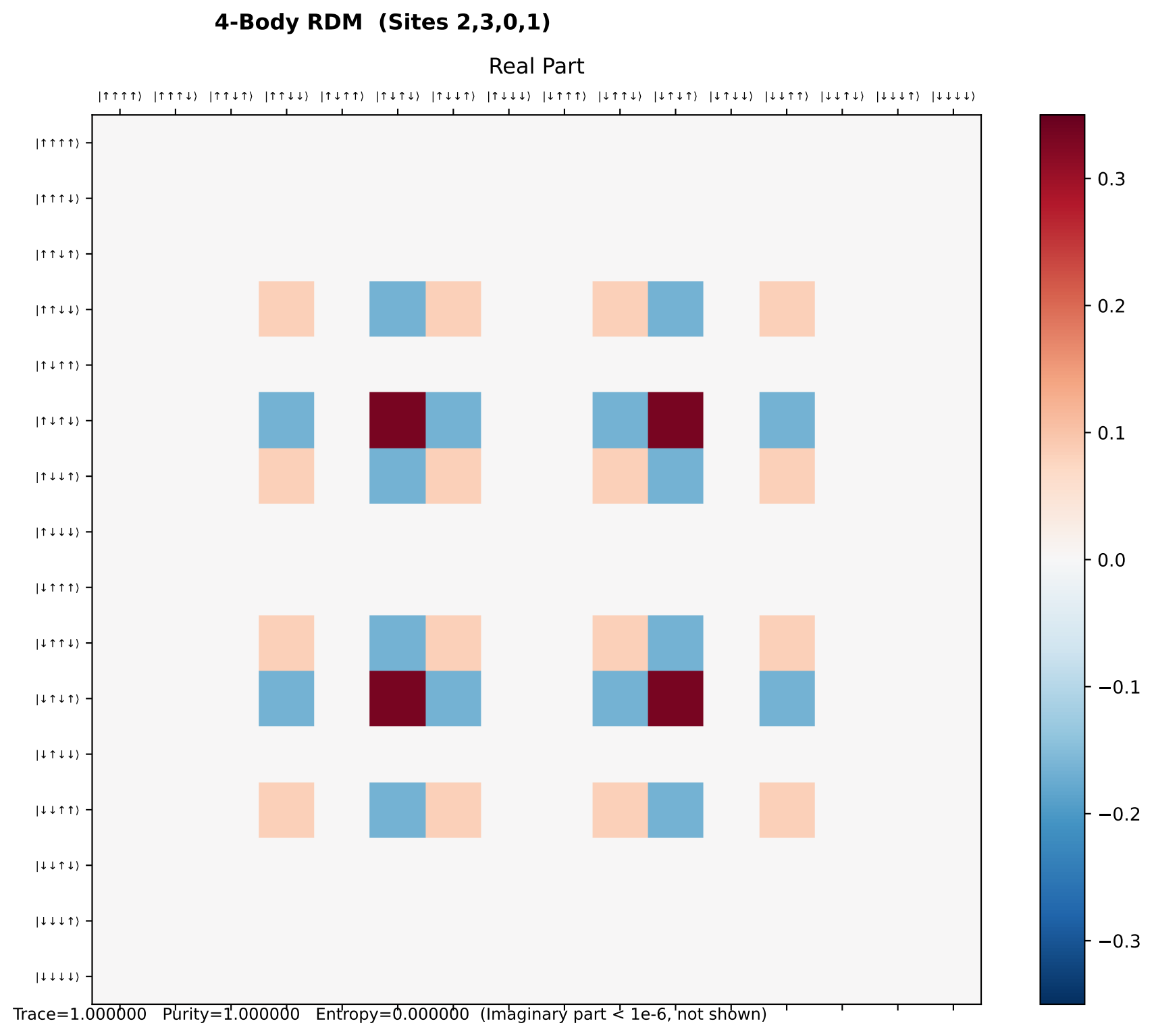
Each spin probed as a general superposition
|phi_k> = cos(t/2)|up> + exp(ip)*sin(t/2)|down>

Method	continuous_complex
C_LB (basis)	0.66666667 <- discrete search
C_LB (cont.)	0.66666667 <- continuous search
Gain	-0.00000000

Continuous Optimization Details
==========

Parameters per spin: theta [0,pi], phi [0,2pi]
Total search space : 4 x n = 16 parameters

Best thetas psi1:
[3.142, 0.0, 3.142, 0.0]
Best thetas psi2:
[0.0, 3.142, 0.0, 3.142]


4-Body RDM (Sites 2,3,0,1)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.216130e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 0, 1)	0.50000000
(2, 3)	(0, 1)	0.58333333
(2, 0)	(3, 1)	0.33333333
(2, 1)	(3, 0)	0.58333333
(2, 3, 0)	(1,)	0.50000000
(2, 3, 1)	(0,)	0.50000000
(2, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME (exact eq.1) = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.
lower bound [eqs.6-9]

C_LB = 0.66666667
Gap = 0.24620426 (C_exact - C_LB)
Best z1 = [0, 1, 0, 1]
Best z2 = [1, 0, 1, 0]

Continuous Optimization
==========

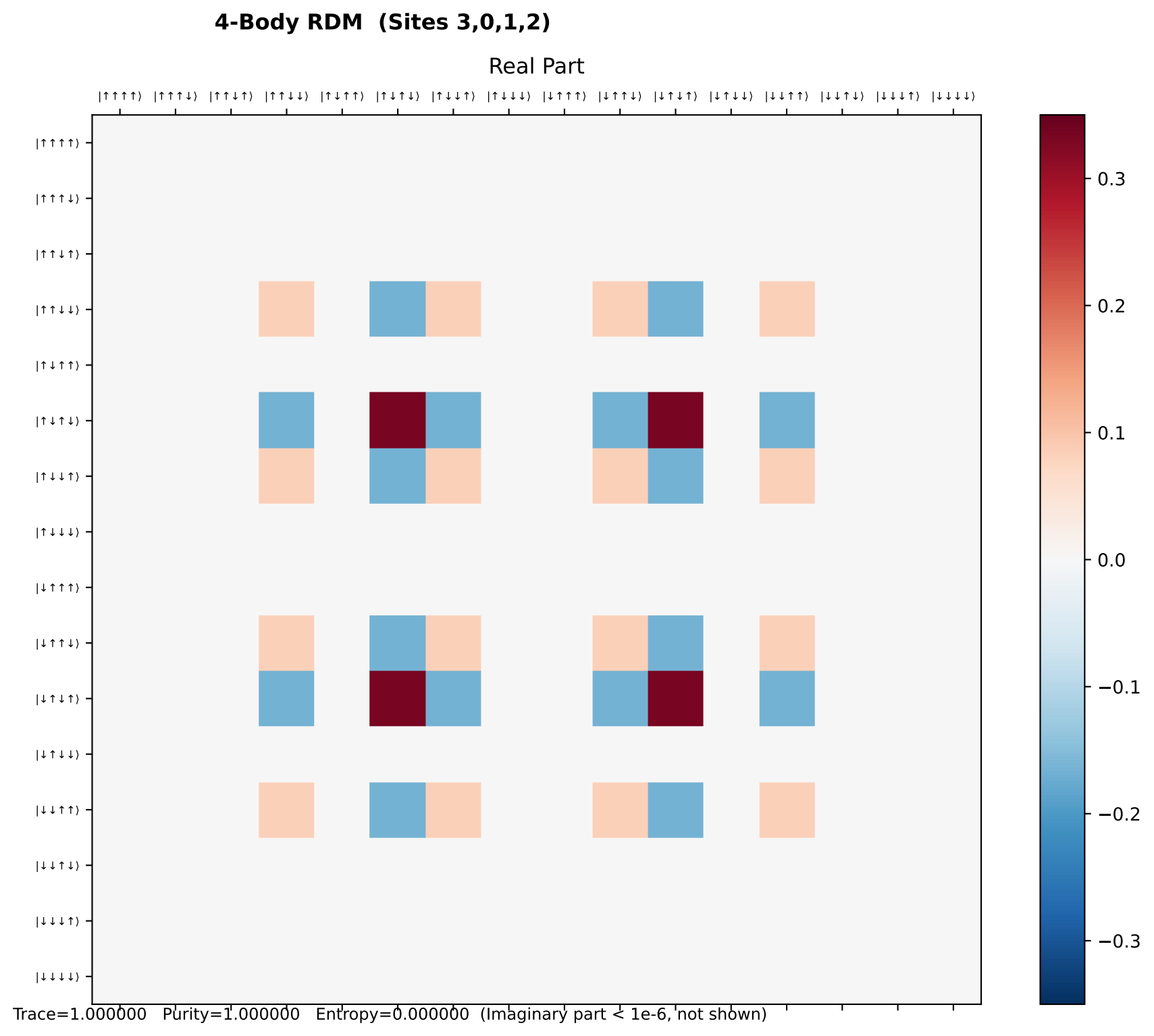
Each spin probed as a general superposition
|phi_k> = cos(t/2)|up> + exp(ip)*sin(t/2)|down>

Method	continuous_complex
C_LB (basis)	0.66666667 <- discrete search
C_LB (cont.)	0.66666667 <- continuous search
Gain	-0.00000000

Continuous Optimization Details
==========

Parameters per spin: theta [0,pi], phi [0,2pi]
Total search space : 4 x n = 16 parameters

Best thetas psi1:
[3.142, 0.0, 3.142, 0.0]
Best thetas psi2:
[0.0, 3.142, 0.0, 3.142]


4-Body RDM (Sites 3,0,1,2)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 3,0,1,2)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.380952e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(0, 1, 2)	0.50000000
(3, 0)	(1, 2)	0.58333333
(3, 1)	(0, 2)	0.33333333
(3, 2)	(0, 1)	0.58333333
(3, 0, 1)	(2,)	0.50000000
(3, 0, 2)	(1,)	0.50000000
(3, 1, 2)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME (exact eq.1) = 0.91287093
C_GME > 0: genuine entanglement confirmed across ALL cuts.
lower bound [eqs.6-9]

C_LB = 0.66666667
Gap = 0.24620426 (C_exact - C_LB)
Best z1 = [0, 1, 0, 1]
Best z2 = [1, 0, 1, 0]

Continuous Optimization
==========

Each spin probed as a general superposition
|phi_k> = cos(t/2)|up> + exp(ip)*sin(t/2)|down>

Method	continuous_complex
C_LB (basis)	0.66666667 <- discrete search
C_LB (cont.)	0.66666667 <- continuous search
Gain	-0.00000000

Continuous Optimization Details
==========

Parameters per spin: theta [0,pi], phi [0,2pi]
Total search space : 4 x n = 16 parameters

Best thetas psi1:
[3.142, 0.0, 3.142, 0.0]
Best thetas psi2:
[0.0, 3.142, 0.0, 3.142]